

GHANA COMMUNICATION TECHNOLOGY UNIVERSITY



FACULTY OF COMPUTING AND INFORMATION SYSTEMS

FINAL YEAR UNDERGRADUATE PROJECTS MANUAL

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1. PURPOSE OF THE PROJECT COURSE

The courses in the BSc degrees at GCTU focus on developing competencies in a specific area. The project course however utilizes all of the skills from these specific areas to solve one real-world problem. Computing professionals provide solutions by considering both technical and business perspectives, ensuring that the chosen approach meets the problem's needs and constraints. Specifically, this course builds upon common concepts explored in previous system development courses.

1.1. Aims and Goals

The course aims to give students the experience of working in small groups and participating in all the development phases (requirements analysis, design, construction, testing, and documentation) of a nontrivial software system. It also gives valuable teamwork experience.

The project assesses your ability to select a problem, analyze and propose solutions then design, build, and test a software system.

It tests your ability:

- to make good decisions and recover from bad ones
- to utilize the collective skill set of the team to overcome various types of difficulties that may emerge
- to work within the constraints of limited resources, particularly time
- to document design decisions and communicate technical concepts

1.2. Course Delivery

Groups of students will liaise with an academic supervisor. Presentations will be conducted on selected topics from Information Technology, Computer Science, Information System and Pervasive and Mobile Computing. The course will be delivered using weekly consultation sessions with student groups (refer to course's timetable for more details). Assessments will take the form of attendance at meetings, written deliverables, presentations and course participation.

1.3. Learning Outcomes and Objectives

While no two projects will be the same, all projects will test students in the same way such that upon the successful completion of this course, students will be able to:

1. Prepare a project proposal for a technology-based project following the standard system development methodology.
2. Make and defend sound software design and development decisions.
3. Manage a project from start to finish meeting stated milestones and timelines
4. Demonstrate interpersonal skills, teamwork, and effective use of appropriate technology associated with system planning and development.
5. Demonstrate effective written and oral communication techniques through the assessment components (detailed in later sections).
6. Discuss and evaluate their contribution to a group project.
7. Review and provide feedback on other group projects.

2. PROJECT DESCRIPTIONS

2.1. Project Scope

The scope of an undergraduate final-year project should clearly define the problem or opportunity being addressed. The extent to which a software solution is fully developed depends on the type of project being undertaken.

2.2. Types of Projects

Students may select a project idea that falls within one of the following four categories:

- Technical Computer Science or Information Technology Projects

2.2.1 Technical Projects

A technical project focuses on producing clean, well-factored, error-free code in a software solution that addresses a particular problem. This type of project should produce a good (best) software solution to a problem but does not have to prove it is the best solution.

A technical project may still apply to a particular domain and have practical uses. A good breadth of algorithmic knowledge is imperative. Students attempting this type of project should have the necessary algorithmic programming skills in the area of application and should have reasonably good knowledge of related areas as well.

3. MANAGING PROJECT EXECUTION

Project management deals with organizational matters that are needed for effective teamwork. An identified group leader can take the lead in the coordination of project activities, or the group can decide to share coordination responsibilities. Project management includes the following:

- Organizing group meetings and keeping track of deadlines
- Managing shared resources (such as documentation, software repository, etc.)
- Integrating individual contributions coherently, and resolving ambiguities and conflicting information including collating different sections into a project document representing the work of the entire group
- Ensuring that the project document is proofread, and collated to ensure consistent layout, font styles, sections, and language style
- Integrating different parts of the program into one software system
- Running integration tests and ensuring that the whole system works as intended
- Fixing software bugs that result from incompatibilities between software modules contributed by different group members.
- Other matters relating to the group's dynamics.

3.1 Group Formation and Size

Students are expected to choose their group members from amongst the students registered for this course. A group size of at least two (2) or three (3) is recommended. However, exceptions may be made for large, complicated projects, pending approval from the Head of Department or Project Coordinator. Group sizes exceeding 4 will not be permitted.

3.2 Group Member Roles and Contributions

Project groups often function more effectively when group members have designated roles. Three core roles and responsibilities are:

- **Project Lead:** responsible for keeping the group on task, distributing the workload, meeting deadlines, and ensuring smooth group communication and coordination internally as well as accountability with the course coordinator and project requirements.
- **Documenter Lead:** responsible for recording group discussions and decisions, documenting various aspects of the project's progress, and ensuring well-formed reports and project documents are produced.

- **Development Lead:** responsible for overseeing the collaborative software aspects of the project, troubleshooting major technical problems related to code or integrative software, and ensuring code reviews, merges and testing are done well.

The entire project team should be engaged in discussions, documentation, and development of the project. All members are expected to contribute towards the project.

Groups will also have the option to rotate the roles among members for different stages of the project. This will allow members to gain experience through being responsible in different areas of the project management.

3.3 Project Selection and Proposal

A list of available project ideas will be made available through the project coordinator or the Head of Department. Students are advised to start exploring their options early and to select a suitable idea that is of interest by the designated date. Alternatively, students may propose an idea of their own. Proposed project ideas require approval by the Head of department or project coordinator. Only ideas that have been approved can proceed further.

3.4 Project Supervision

Projects will be supervised by an assigned supervisor within the faculty. A Teaching assistant will be available for closer consultation, assistance and one-on-one advice with groups. A co-supervisor (an identified university staff member) may be assigned for selected projects; based on the nature of the project and the availability of resources.

3.5 Software Development

Groups are encouraged to adopt agile software development methods as the practice is relatively simple to adopt and provides greater project visibility, predictability, and flexibility.

It also encourages the development of high-performance teams and early prototypes. A GitHub repository is required for each group project to facilitate project management and keep track of software and design changes. The repository will also be used by the coordinator, research supervisor and teaching assistants to monitor, review and assess components of each project throughout the course. Note: students should create their own GitHub accounts if they do not have one.

3.6 Presentations and Demonstrations

Interim presentations are intended to be formative assessments: demonstrations and brief status reporting will enable students to receive feedback on their work. The interim presentations will also be used to give and receive peer feedback. Peer reviews/critiques are intended to encourage lateral thinking and the development of review skills. Marks will be awarded to encourage participation in these activities.

4. PROJECT ASSESSMENT AND GRADING

A detailed breakdown of the assessment components and their weightings follow below:

PROJECT REPORT (50)

INTRODUCTION/ PROJECT CONTEXTUALISATION Background/Problem Specification: Very clear specification of the problem with reason and purpose of the research unambiguously stated. Clear aim and objectives.	7
Scope/ Significance of study: Acknowledgement of relevant boundaries, limitations to scope of study	3
MAIN BODY OF WORK Literature Review: Comprehensive and critical review of literature demonstrating breadth and depth. Identification and analysis of key issues and arguments within the research field structured as a coherent and sustained discussion of relevant material. Adequate recognition of sources collated in References and Bibliography sections as appropriate using Harvard style.	10
System Specification & Design: Clear description of the conceptual model and process methodology that were used for the successful development of the system. Use of appropriate information gathering techniques and system development tools. Demonstration of good system analysis, accurate system and design requirements specification, and appropriate specification of software and hardware requirements.	10

Document Presentation: Lucid, concise and accurate use of language. Presented in a manner consistent with the style of an academic journal. Visually stimulating where appropriate.	3
Reference: Correctly presented list of reference/ bibliography in line with Harvard style.	3

SOFTWARE ARTIFACT (30)

Interface Design	15
Business Logic	8
Error Handling	7

PROJECT PERFORMANCE REPORT (20)

Appreciation and comprehension of task in hand	5
Understanding of the appropriate process methodology, tools and techniques	5
Diligence and competence in performing the task without excessive help	5
Evidence of sustained activity and single mindedness of purpose throughout the period	5

4.3. Final Project Report

The final project report aggregates the interim reporting with additional information about the project. A detailed description of the report format and content is included below.

A. PRELIMINARY SECTION

1. Title Page

This is made up of the name of the university, logo, name of faculty and department to which it is submitted, title of the project work, in partial fulfillment of requirement for what degree/diploma and by (full name of author(s)), supervisors name and Year of presentation (centered). A specimen is presented in Appendix A.

2. Certification Page

A project must be authenticated by a declaration as being the previously unpublished work of the

author. In this regard, students are to ensure that there are no indications of plagiarism in the project as it would lead to disqualification and rejection of the project (plagiarism is the act of taking somebody's work and passing it off as one's own).

The declaration should have the rendition shown in Appendix B. It consists of the student's declaration and certified by the supervisor and the Head of the Department who append their signatures and by so doing affirm that in their opinion the project work:

- i. is an independent, original contribution to the knowledge of the subject concerned of which.
- ii. literary presentation is satisfactory.

3. Abstract

This is a brief summary of the project work and is most likely to be widely published and read. It should have a concise description of the Problem(s) addressed, the Methods for the solution, the Results, and the Conclusions. The whole abstract should be composed as one paragraph. A maximum limit is set for half a page or 250 words.

4. Table of Contents

This highlights the chapters and subchapters and the contents of the material within the covers of the project work including the pages where they are located. A specimen is presented at Appendix C.

5. List of Tables, List of Figures and List of Abbreviations

These should be on separate pages and included only as appropriate.

6. Prefatory Material

Where appropriate, material pertaining to the Preface, Foreword and Acknowledgements etc. could be presented on separate pages. However, the Acknowledgement page is mandatory.

B. MAJOR COMPONENT OF THE PROJECT WORK

7. Text

This comprises the main Chapters (5 chapters are allowed for both Diploma and Degree) and Subchapters (divisions and subdivisions) and their namely: Introduction, Literature Review (may include Theory), System Specification and Design, System Implementation, Conclusion and Recommendation. A specimen is presented at Appendix C.

C. WRITING MECHANICS

8.1. Paper and Production Processes

The paper type and format to be used for the project work should be International standard paper size A4, dimensions (297x210 mm). The final project work material must meet high standards of permanence, legibility, uniformity and reproducibility.

The first-deposit copies will not be subject to paper or print specifications, but must be legible and follow all formatting guidelines prescribed by this manual. Two (2) copies of final project

report are to be book-bound, with type on one side of the page only.

8.2. Type Size and Style

The same 12-point typeface (Times New Roman font) is to be used for all preliminary pages, tables and figure captions, text, appendices, references and page numbers except for Footnotes/ Endnotes when other distinguishing font types may be used.

8.3. Spacing •

Text of the project work must have 1.5 spacing. The same should apply in between paragraphs. However, for the Table of Contents, Footnotes, Quotes in the text and bibliography, single-spaced formatting may be used.

8.4. Captions/Legends for Tables and Figures

Captions for tables should be numbered and reflect the content of the table and set above the table. In the case of figures, the captions should be related to the contents and set below the figure. Numbering for the Tables and Figures should be linked to the chapters i.e. numbers for the Tables and Figures should begin with the number of the chapter where it is sited. Captions for tables should be up i.e. above tables and that of figures down i.e. below figure.

8.5. Pagination

a. All pages must be numbered consecutively. Each Chapter should begin on a separate page. The University requires that page numbers be centered two lines below the bottom margin.

Title page

b. On the title page, the page number is omitted, though it is considered as page "i".

Front Matter (Preliminary Pages)

c. Preliminary pages are numbered with lower case Roman numerals, beginning with (ii) following the title page centred at the bottom. This includes the first page of each back matter chapter/section, (i.e., appendices, endnotes etc.).

Reference and other Matter

d. Pagination for the References and other material is by Arabic numerals.

Facing Pages

e. Printing should be one-sided i.e. on facing pages only.

8.6. Length of Project Work

It is recommended that the text number ranges between 30 to 50 pages for Diploma and 40 to 60 pages for Degree.

Paragraphing

First line of paragraphs must be indicated by indentation. A Paragraph should not be less than five lines and not more than eight lines. Each paragraph should develop one main idea and should have a topic sentence which expresses this idea. Paragraphs should be right "justified".

Margins

Margins should have the following minimum settings:

Top—2.5 cm; Bottom—2.5cm; Left —4.0 cm; Right —2.5 cm. *Exceptions:* A larger top margin may be used on pages with main headings, a larger bottom margin on pages that otherwise would have just a heading at the bottom of the page, larger margins around tables and figures, and smaller margins on appendix

APPENDIX A - Sample Title Page for Project Work

GHANA COMMUNICATION TECHNOLOGY UNIVERSITY (GCTU)



FACULTY OF COMPUTING AND INFORMATION SYSTEMS (FOCIS) (15)

[...DEPARTMENT NAME...]

TITLE:

[TITLE OF PROJECT]

(15)

A Project Work Submitted in Partial Fulfillment of the Requirements For
BSc. /Diploma [NAME OF DEGREE]

(12)

BY:

[LIST OF NAMES OF STUDENTS]

SUPERVISOR:

[NAME OF SUPERVISOR]

[MONTH, YEAR]

APPENDIX B - Sample of Declaration

DECLARATION

This project is presented as part of the requirements for BSc. /Diploma [name of program] awarded by Ghana Communication Technology University. I hereby declare that this project is entirely the result of hard work, research, and enquires. I am confident that this project work is not copied from any other person. All sources of information have however been acknowledged with due respect.

AUTHOR: XXXXXXXX
STUDENT ID: XXXXXXXX

SIGNATURE.....
DATE:

AUTHOR: XXXXXXXX
STUDENT ID: XXXXXXXX

SIGNATURE.....
DATE:

AUTHOR: XXXXXXXX
STUDENT ID: XXXXXXXX

SIGNATURE.....
DATE:

SUPERVISOR: XXXXXXXX

SIGNATURE.....
DATE:

HOD: XX

SIGNATURE.....
DATE:

APENDIX C – Table of Contents of final report

TABLE OF CONTENTS

Title Page

Declaration i

Abstract ii

Table of Contents iii

List of Tables iv

List of Figures v

List of Abbreviation vi

Acknowledgement vii

Chapter One: Introduction

1.1 Background to Study

1.2 Statement of the Problem

1.3 Aim & Objectives

1.4 Significance of the Study

1.5 Organization of the Study

1.6 Conclusion

Chapter Two: Literature Review

2.1 Overview of Relevant Literature

2.2 Theoretical Framework

2.3 Previous Studies and Related Works

2.3.1 Previous Studies and Related Works 1

2.3.2 Previous Studies and Related Works 2

2.3.3 Previous Studies and Related Works 3

2.3.n Previous Studies and Related Works n

2.4 Summary of Literature Review

2.5 Conclusion

Chapter Three: System Specification and Design

3.1 Methodology

3.2 System Specification

3.2.1 Analysis of Requirements

3.2.2 Stakeholder Analysis

3.2.3 Functional Requirements

3.2.4 Non-functional Requirements

3.2.5 Use Cases

3.3 System Design

3.3.1 Architectural Design

3.3.2 Database Design

3.3.3 User Interface Design

3.3.4 Security Considerations

3.4 Conclusion

Chapter Four: System Implementation

4.1 Development

4.1.1 Programming Languages and Tools Used

4.1.2 Implementation Details

4.1.3 Testing and Quality Assurance

4.2 Evaluation

4.2.1 Evaluation Metrics

4.2.2 User Testing

4.2.3 Performance Evaluation

4.3 Conclusion

Chapter Five: Conclusion and Recommendations

5.1 Discussion

5.1.1 Comparison with Existing Systems

5.1.2 Challenges Faced

5.1.3 Lessons Learned

5.1.4 Contributions to the Field

5.1.5 Future Work

5.2 Conclusion

References /Bibliography

Appendices

Appendix A. Code Snippets

Appendix B. Screenshots

Appendix C. User Manuals

Compilation of Bibliography/References

You will certainly be using books, journal papers, articles, conference papers, URLs and so forth, to provide you with information to do your project. In addition, the background section of your project will discuss other people's work in the field.

You must acknowledge all your information sources. This is done by providing a list of references at the back of your project report, and referring to (i.e. citing) the individual items in this list at the appropriate points in your project report.

The reference list is usually put at the very end of project reports, after all the chapters and appendices. It is not usual to give a section number to the references. Also if your reference list is trivial/non-existent, it is an indication that you have not really delved into literature and so would almost certainly affect your grade.

The Harvard system of referencing

There are various ways of listing the references. If you look in the back of research papers and some textbooks, you will see examples. The *Harvard system* is an *author-date system*. For example, if reference is to a contribution from a book the following is acceptable:

For First Edition:

Surname, initials (Date) *Title of book*, Publisher's name, Place of publication

For book other than First Edition:

Surname, initials (Date) *Title of book* (? edn), Publisher's name, Place of publication

For a journal article, the following details are expected:

Surname, initials (Date) 'Title of article', *Journal name*, volume number part number, pages

Here are examples:

Armstrong, JM., and Mitchell, RJ (1994) 'Uses and Abuses of Inheritance', Software Engineering Journal, 26:1, 19-26

Communications of the ACM (1995), 'Special Issue on Cognition and Software Development', 38:6

Cargill, TA (1990) 'Does C++ really need multiple inheritance', Proceedings of UK Unix User Group Conference, 53-59

Goguen, JA and Meseguer, J (1989) 'Order-sorted algebra I: equational deduction for multiple inheritance, overloading, exceptions and partial operations', Technical Monograph PRG-80, Oxford Polytechnic Computing Laboratory, Programming Research Group.

If reference is to a contribution from the internet this is how it is cited:

Inheritance -- multiple and virtual inheritance[online]. Available:

<http://yosefk.com/c++faq/inheritance-multiple.html>[accessed 3 September 2008]

The reference list is in alphabetical order. Note that each individual reference needs to be this detailed to enable a reader of your project report to find it. You must, therefore, give the author(s), the title, when it was published, where it was published (if a journal article, then state the particular volume and issue number, if a book, state the name of publisher), and (also if a journal article), which page numbers the article covers.

Items in a Harvard reference list, such as the one shown above, are cited in the main body of your dissertation with the appropriate *author-date* style. Here are some sample citations, using this style: *A recent survey on software development in the communications of the ACM [Ayisi 2012] reports that the choice of programming language apparently does not have any appreciable effect on levels of software reuse. Maybe one of the greatest dangers that exist is that object-oriented languages (such as C++) are now so popular, they are being used to develop software for safety critical applications before we have any formal basis for inheritance between classes in these languages [Agyei 2010]. Several attempts have been made to formalize types of inheritance, particularly multiple inheritance [Addo 2011], [Smith 2013], though none has been widely accepted as correct.*

Remember that every item in your list of references must be cited (referred to) at least once somewhere in the text. If you wish to list any literature that gives general background or is helpful in some other way but is not cited in the report, then this list goes into a separate section called the Bibliography.

4.4. Final Presentations and Posters

Final presentations and posters typically mark the culmination of a project or academic endeavor. Here's a general overview of what each might entail:

Final Presentations:

Purpose: Final presentations are intended to showcase the findings, outcomes, or progress of a project or research endeavor to an audience. This audience could include peers, supervisors, stakeholders, or the public.

Content: Presentations often include an introduction to the project, its objectives, methodology, key findings, analysis, and conclusions. Depending on the context, it might also include recommendations, future directions, or implications of the work.

Format: Presentations can vary widely in format, ranging from formal PowerPoint presentations to more interactive formats like demonstrations or panel discussions. The format should suit the audience and the goals of the presentation.

Delivery: Presenters should aim to engage the audience effectively, communicate clearly, and address any questions or concerns that arise. Practice and preparation are essential for a successful presentation.

Posters:

Purpose: Posters serve as visual summaries of a project or research study, typically displayed at conferences, symposiums, or other events. They provide an opportunity for presenters to share their work in a concise and visually appealing format.

Content: Posters typically include sections such as the title, abstract, introduction/background, methodology, results/findings, conclusions, and references. Visual elements such as graphs, charts, images, and diagrams are commonly used to illustrate key points.

Design: Effective poster design is crucial for capturing viewers' attention and conveying information clearly. Posters should be visually appealing, easy to read from a distance, and organized in a logical manner.

Presentation: Presenters may stand by their posters during designated sessions to answer questions, provide additional context, or engage with attendees interested in their work. It's essential to be prepared to discuss the poster's content in-depth and communicate its significance effectively.